About This Dashboard

What This Dashboard Shows

Page 1 – Risk Overview

Displays overall default rate, FICO impact, and risk distribution.

Page 2 – Credit & Income Analysis

Explores how income bands, DTI tiers, and credit history depth relate to default risk.

Page 3 – Borrower Risk Profile

Shows FICO vs interest rate patterns and loan purpose risk segmentation.

Page 4 – Model Performance

Displays confusion matrix, Recall, False Negative Rate, and threshold sensitivity.

How to Use the Dashboard

- Use slicers on the left to filter by:
  - Credit Policy (0 = did not meet criteria, 1 = approved)
  - Loan Purpose
  - FICO Band
- All visuals update dynamically.

Colour Convention

- **Red = High Risk / Default**
- **Grey = Fully Paid**
- Interest rate (int.rate) is displayed as an annual percentage.

Important Statistical Note

Approximately **16.01%** of loans in this dataset defaulted.
Performance metrics are evaluated on this minority class to ensure high detection of defaulting borrowers.

9,578
Total Loans

1533
Total Defaults

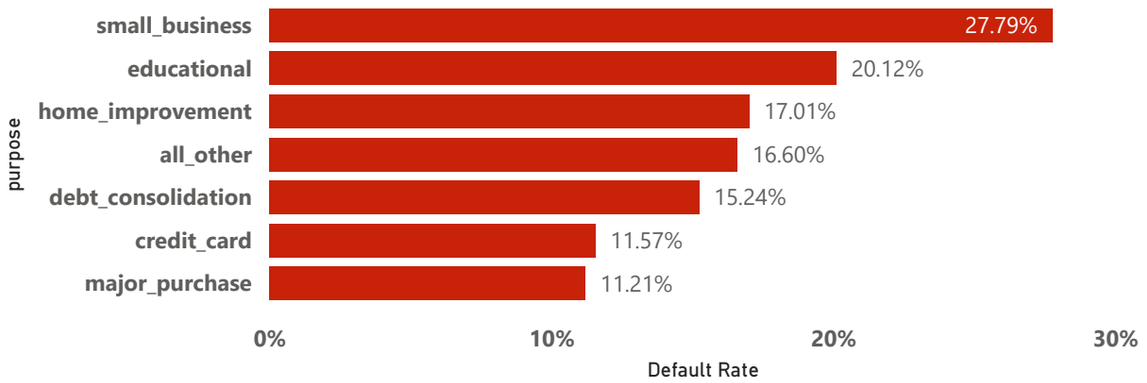
Minority Class = 16.01%

16.01%
Default Rate

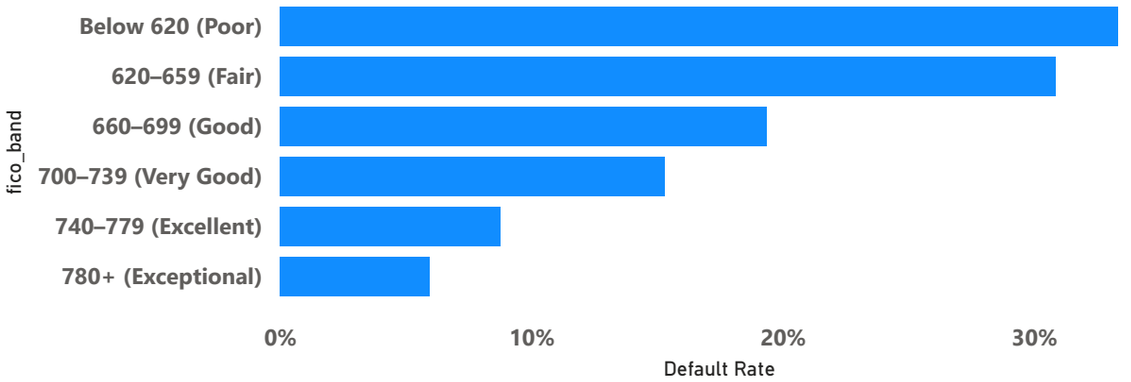
698
Avg FICO Default

713
Avg FICO Paid

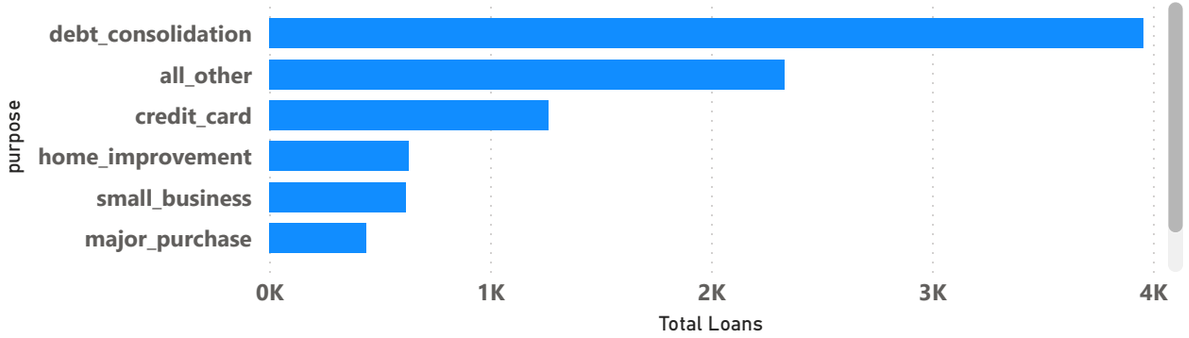
Default Rate by Purpose



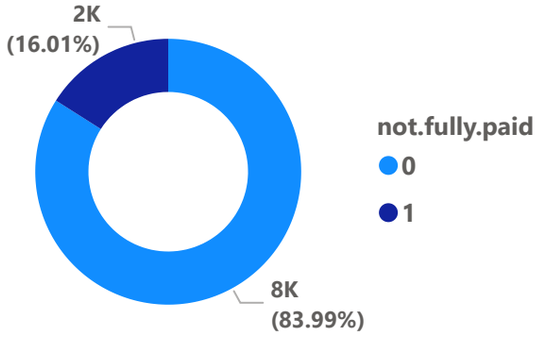
Default Rate by fico_band



Total Loans by purpose

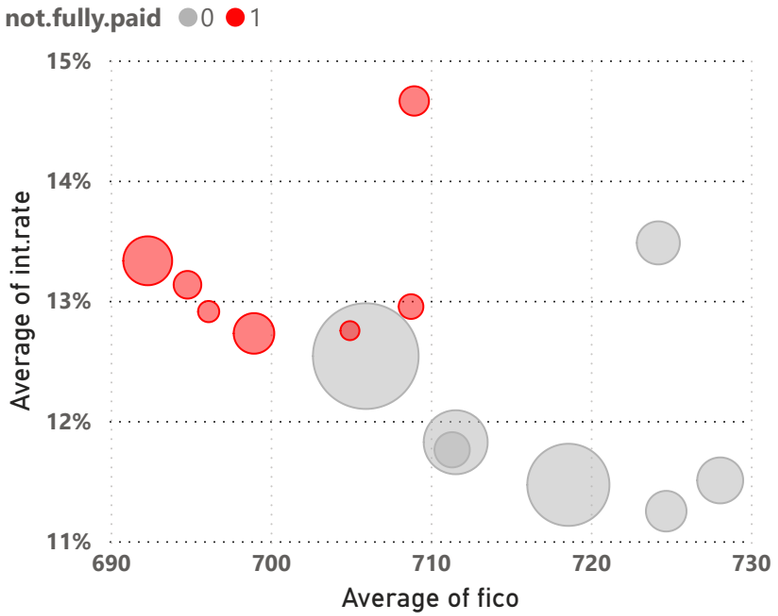


Total Loans by not.fully.paid

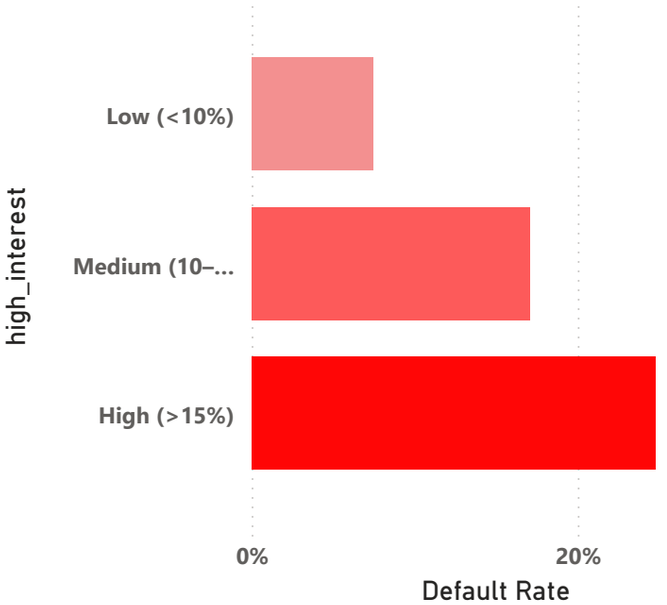


Default risk drops consistently as credit quality improves, confirming FICO as a primary predictor.

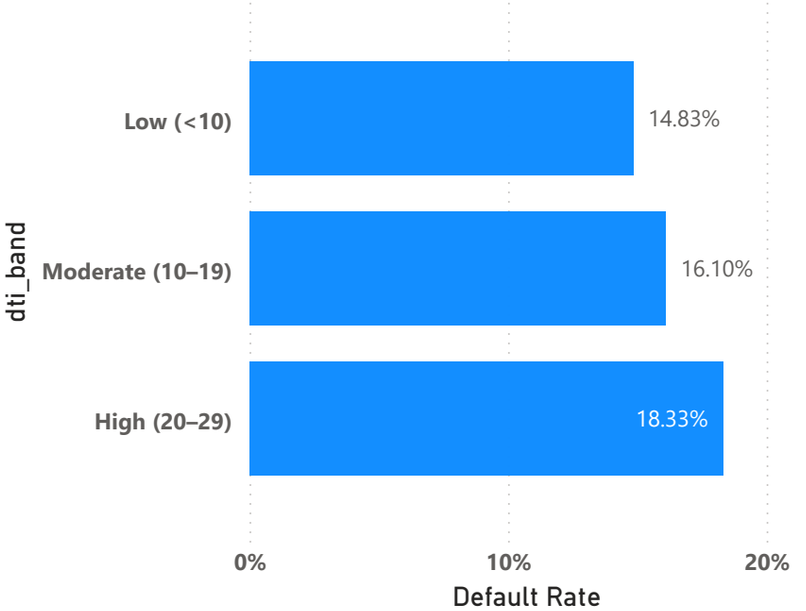
FICO Score vs Interest Rate



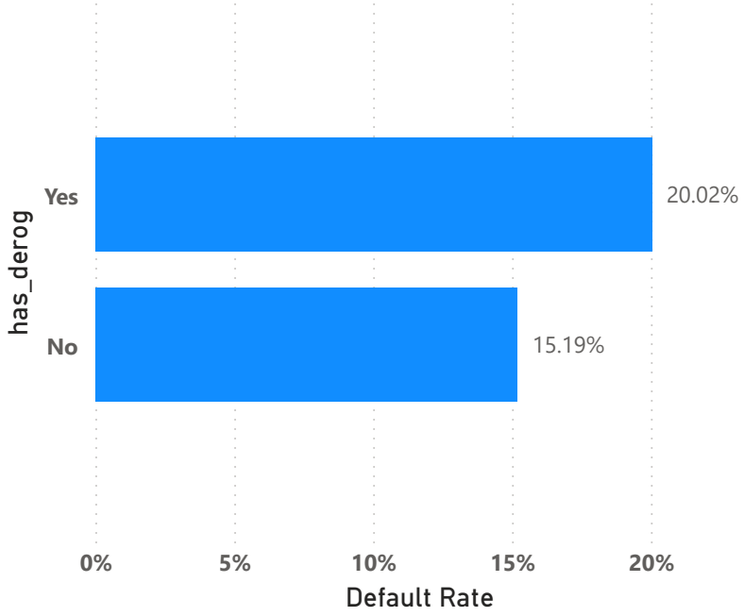
Default Rate by Interest Rate Tier



Default Rate by dti_band



Default Rate by has_derog



13.20

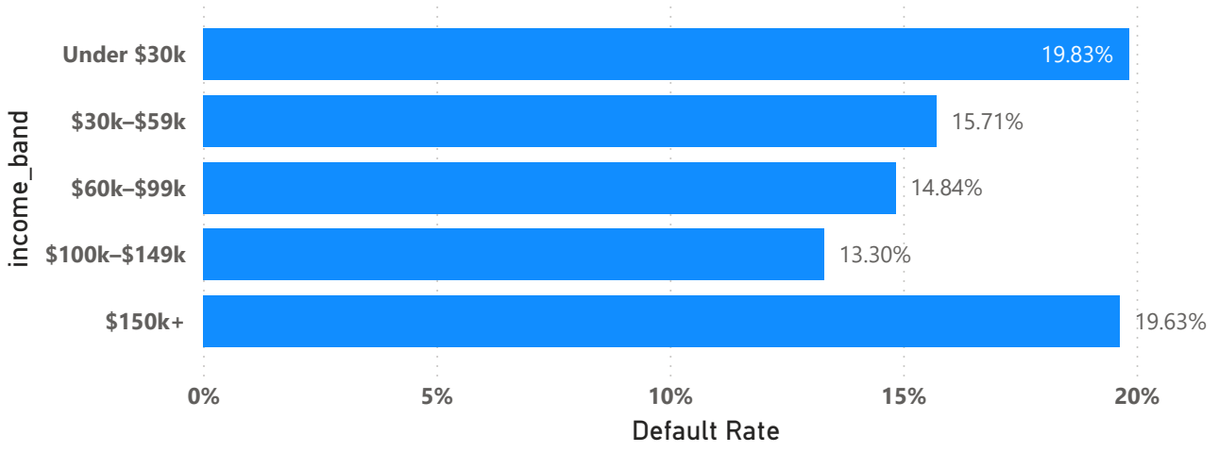
Avg DTI Default

12.49

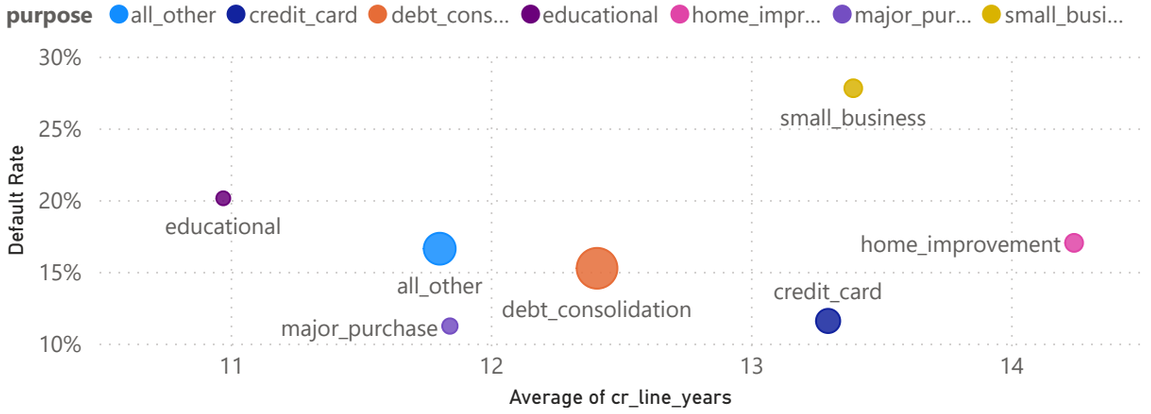
Avg DTI Paid

Borrowers with derogatory records default at nearly double the rate.

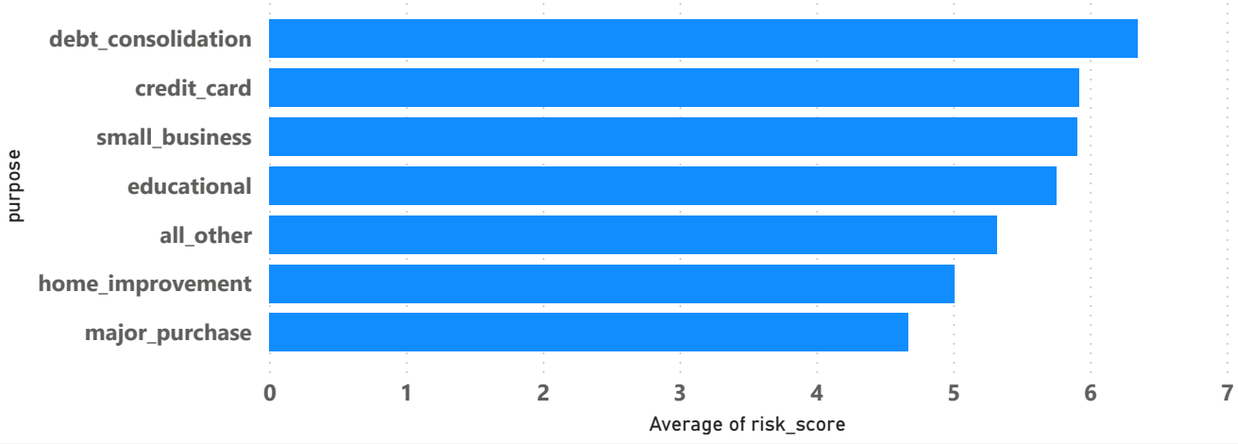
Default Rate by income_band



Credit Line Years vs Default Rate by Purpose



Average Risk Score by Loan Purpose



fico_band	Total Loans	Total Defaults	Default Rate	Avg Int Rate	Avg DTI
Below 620 (Poor)	3	1	33.33%	0.18	7.77
620–659 (Fair)	486	150	30.86%	0.15	14.05
660–699 (Good)	3,732	723	19.37%	0.14	13.76
700–739 (Very Good)	3,127	478	15.29%	0.12	13.08
740–779 (Excellent)	1,709	150	8.78%	0.10	10.49
780+ (Exceptional)	521	31	5.95%	0.09	7.11
Total	**9,578**	**1533**	**16.01 %**	**0.12**	**12.61**

credit.policy

0

1

View Approved Loans Only

purpose

All

fico_band

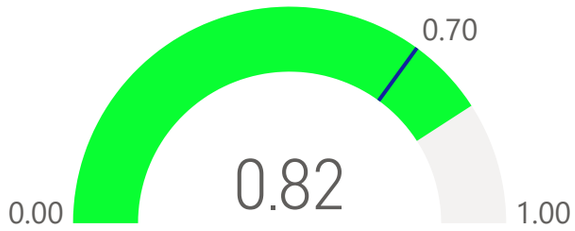
- 620–659 (Fair)
- 660–699 (Good)
- 700–739 (Very Good)
- 740–779 (Excellent)
- 780+ (Exceptional)

True Positives	False Negatives	False Positives	True Negatives
1253	280	5076	2969

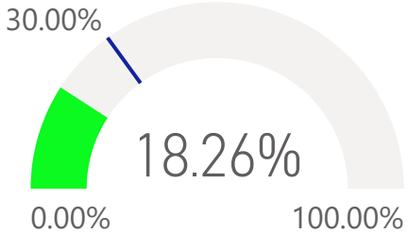
81.74%

Recall

Recall (Target ≥ 70%)



False Negative Rate (Target ≤ 30%)

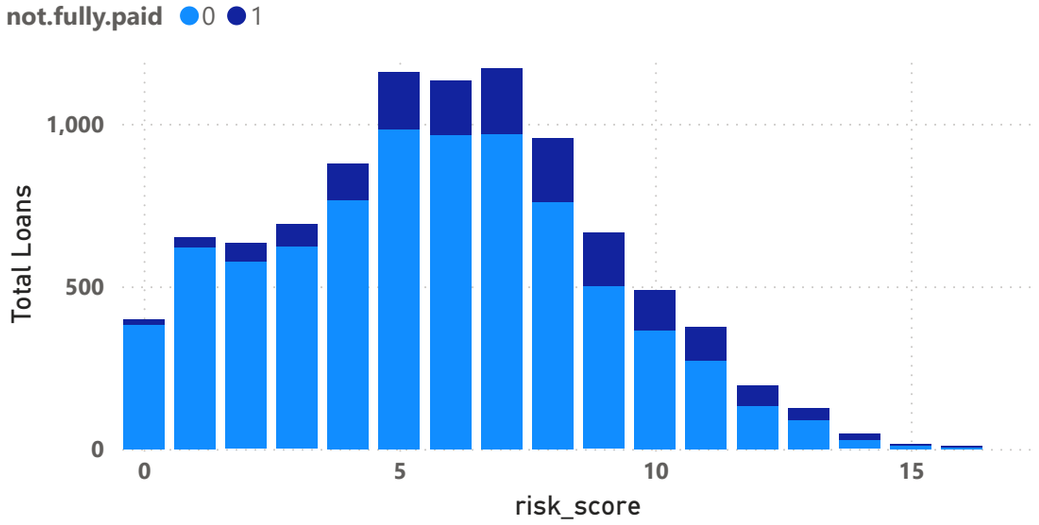


18.26%

FNR

not.fully.paid	0	1
0	2969	5076
1	280	1253

Total Loans by risk_score and not.fully.paid



The model misses 18.26% of actual defaults, staying well below the 30% threshold required by management.